

BENJAMÍN I. FORTUÑO

[linkedin.com/in/benjamin-fortuno](https://www.linkedin.com/in/benjamin-fortuno) • benjaminignacio.fortuno@polimi.it • (+39) 344-4276207

PhD Researcher – Surgical Robotics & Robot Learning.

Learning-based autonomy for microsurgical manipulation, integrating control theory, imitation/reinforcement learning, and vision-based perception. Currently deploying real-robot visuomotor policies on the **Symani microsurgical robot** in collaboration with **Medical Microinstruments (MMI)**.

EDUCATION

Politecnico di Milano

2024 – Present

Ph.D. Bioengineering (Co-funded by Medical Microinstruments) — Milan & Pisa, Italy

- Architect full training stack (ROS2, Isaac Sim, multi-camera perception, datasets)
- Safety-aware control under contact constraints for visuomotor policies
- Teaching Assistant — Medical Robotics & Computer-Aided Surgery Laboratory (Prof. Elena De Momi)

Politecnico di Milano

2022 – 2024

M.Sc. Biomedical Engineering — Milan, Italy

- Thesis: Closed-loop adaptive controller with electromagnetic tracking for autonomous transesophageal echocardiography
- 1st place Hospital-based Health Technology Assessment (Humanitas)

Pontificia Universidad Católica de Chile

2018 – 2022

B.Sc. Electrical Engineering (Automation & Robotics Minor) — Santiago, Chile

- Teaching Assistant (3 years): Linear Algebra, Calculus, Introduction to Robotics, Advanced Signal Processing

PUBLICATIONS

- X. Zhang, **B. I. Fortuño Jara**, M. Di Mauro, V. Cannizzaro, A. Peloso, A. Bicchi, et al. "Collaborative control framework of a robotic transesophageal echocardiography system for guiding structural heart interventions." *IEEE/ASME Transactions on Mechatronics*, 2026.
- A. Bicchi, X. Zhang, **B. I. Fortuño Jara**, V. Cannizzaro, A. Peloso, E. De Momi. "Model-based position control of a tendon-driven variable-length continuum robot for minimally invasive mitral valve repair." *IEEE Transactions on Medical Robotics and Bionics*, 2025.
- X. Zhang, I. Tamadon, **B. I. Fortuño Jara**, V. Cannizzaro, A. Peloso, et al. "Design and hysteresis compensation of a telerobotic system for transesophageal echocardiography." *IEEE Robotics and Automation Letters*, 2024.

RESEARCH & WORK EXPERIENCE

Surgical Robotics Researcher — Medical Microinstruments

Dec 2024 – Present

- Developing **imitation learning** autonomy for the **Symani microsurgical robot**, building an end-to-end pipeline from demonstration capture to real-robot deployment via the **Synaptix platform**
- Added **2 wrist cameras per arm** (Misumi tube cameras) with time-synchronized multi-camera capture, achieving **4 ms** inter-stream sync for high-fidelity demonstration recording
- Deployed learned visuomotor policies via **Synaptix** inference at **50 Hz** closed-loop control, reaching **86%** task success on needle-handling benchmarks using (300 demonstrations of training)

Research Intern — Brain-Machine Interfaces Lab

2020 – 2021

- First/second-level fMRI analysis and multivariate pattern analysis (MVPA) in Python

SELECTED PROJECTS

Autonomous Microsurgical Simulation Framework

- Modular Isaac Sim + ROS2 environment for surgical manipulation research
- Dataset pipelines for imitation and reinforcement learning
- Reproducible evaluation metrics for precision, stability, and success rate

Closed-Loop Adaptive Control for Robotic Echocardiography

- Adaptive controller with EM-tracking feedback
- Stable positioning under anatomical motion and disturbances
- Real-time robust control for surgical guidance

TECHNICAL SKILLS

Programming: Python, C++, C
Robotics/AI: ROS/ROS2, Isaac Sim, OpenCV, PyTorch, RL, IL, Control
Systems: Docker, Git, Linux, MATLAB
CAD/Hardware: Fusion 360, Eagle, PCB design

LANGUAGES

Spanish (Native)
English (Professional)
Italian (Conversational)